1. JavaScript Introduction Theory Assignment

- **Question 1:** What is JavaScript? Explain the role of JavaScript in web development.
  - ➢ JavaScript is a scripting language used to make web pages interactive. It adds dynamic features like animations, form validation, and interactive buttons. It works with HTML and CSS to create responsive and user-friendly websites.
- **Question 2:** How is JavaScript different from other programming languages like Python or Java?
  - ➢ JavaScript runs mainly in browsers, whereas Python and Java run on different environments.
    JavaScript is loosely typed, while Java and Python are more strict.
    JavaScript is mostly used for web development, while the others are used for many fields.
- **Question 3:** Discuss the use of <script> tag in HTML. How can you link an external JavaScript file to an HTML document?
  - ➢ JavaScript can be written or added to an HTML page using the to link external JavaScript.
    This makes code more reusable and clean.

2. Variables and Data Types

- **Question 1:** What are variables in JavaScript? How do you declare a variable using var, let, and const?
  - ➢ Variables store data values.
    We declare them as: var name, let age, const PI = 3.14.
    var is old, let is modern and changeable, const cannot be changed.
- **Question 2:** Explain the different data types in JavaScript. Provide examples for each.
  - ➢ JavaScript has String ("Hello"), Number (25), Boolean (true),
    Undefined (a variable with no value), Null (null),
    Object ({name:"Raj"}), and Array ([1,2,3]).
- **Question 3:** What is the difference between undefined and null in JavaScript?
  - ➢ Undefined means a variable is declared but no value is assigned.
    Null means the value is intentionally empty.
    Undefined happens by default, null is given by the programmer.

3. JavaScript Operators

- **Question 1:** What are the different types of operators in JavaScript? Explain with examples.
  - ➢ Arithmetic:
    + , - , * , / → 5+2.
    Assignment:
    = , += → x = 10.
    Comparison:
    == , === , > → 5 > 3.
    Logical:
    && , || , ! → true && false.

- **Question 2:** What is the difference between == and === in JavaScript?
  - == checks only value.
    === checks value and data type.

4. Control Flow (If-Else, Switch)

- **Question 1:** What is control flow in JavaScript? Explain how if-else statements work with an example.
  - Control flow decides which code will run based on conditions.
    In if-else, if the condition is true, one block runs; otherwise the else block runs.
    Example: if(age>18){…} else {…}.
- **Question 2:** Describe how switch statements work in JavaScript. When should you use a switch statement instead of if-else?
  - Switch checks one value against many cases.
    It runs the matching case block.
    Use switch when you have many conditions for one variable.

5. Loops (For, While, Do-While) Theory Assignment

- **Question 1:** Explain the different types of loops in JavaScript (for, while, do-while). Provide a basic example of each.
  - For loop repeats for a fixed number of times → for(i=0;i<5;i++).
    While loop runs while a condition is true.
    Do-while runs at least once before checking the condition.
- **Question 2:** What is the difference between a while loop and a do-while loop?
  - In while loop, condition is checked first.
    In do-while, the loop runs once before checking.
    So do-while always executes at least once.

6. Functions Theory Assignment

- **Question 1**: What are functions in JavaScript? Explain the syntax for declaring and calling a function.
  - Functions are reusable blocks of code.

```
Function func-name(){
            ..logic
              }
        func-name()
```

- **Question 2:** What is the difference between a function declaration and a function expression?
  - Declaration: function test(){} (hoisted).
    Expression: let fun = function(){} (not hoisted).
    Expressions are stored in variables.
- **Question 3:** Discuss the concept of parameters and return values in functions.
  - Parameters are inputs for a function. Return gives back an output from the function.
    Example: return sum;.

7. Arrays Theory Assignment

• **Question 1:** What is an array in JavaScript? How do you declare and initialize an array?

- An array stores multiple values in one variable.
  Declared as: let arr = [1,2,3].
  We can access items using index numbers.

• **Question 2:** Explain the methods push(), pop(), shift(), and unshift() used in arrays.

- push() adds item at end; pop() removes last item.
  shift() removes first item; unshift() adds item at start.
  These methods help modify arrays easily.

8. Objects Theory Assignment

• **Question 1:** What is an object in JavaScript? How are objects different from arrays?

- Object stores data in key–value pairs.
  Array stores data in index order.
  Objects are used to represent real-world items.

• **Question 2:** Explain how to access and update object properties using dot notation and bracket notation.

- Dot notation: obj.name.
  Bracket notation: obj["name"].
  Both can be used to read or update values.

9. JavaScript Events Theory Assignment

• **Question 1:** What are JavaScript events? Explain the role of event listeners.

- Events are actions like click, mousemove, or keypress.
  Event listeners wait for these actions and run code.
  They make websites interactive.

• **Question 2:** How does the addEventListener() method work in JavaScript? Provide an example.

- It attaches a function to run when an event happens.
  Example: button.addEventListener("click", myFunc);.
  It helps handle multiple events cleanly.

10. DOM Manipulation Theory Assignment

• **Question 1:** What is the DOM (Document Object Model) in JavaScript? How does JavaScript interact with the DOM?

- DOM is a tree structure of the webpage.
  JavaScript can change text, styles, images, and elements.
  It allows dynamic updates on web pages.

• **Question 2:** Explain the methods getElementById(), getElementsByClassName(), and querySelector() used to select elements from the DOM.

- getElementById() selects one element by ID.
  getElementsByClassName() selects all elements with same class.
  querySelector() selects the first element matching a CSS selector.

11. JavaScript Timing Events (setTimeout, setInterval) Theory Assignment

• **Question 1:** Explain the setTimeout() and setInterval() functions in JavaScript. How are they used for timing events?

- setTimeout() runs code after a delay.
  setInterval() runs code repeatedly after fixed time.
  Both are used for timed actions.

• **Question 2:** Provide an example of how to use setTimeout() to delay an action by 2 seconds.

- ```
  setTimeout(function(){
    console.log("Hello");
  }, 2000);
  ```

12. JavaScript Error Handling Theory Assignment

• **Question 1:** What is error handling in JavaScript? Explain the try, catch, and finally blocks with an example.

- Error handling catches and manages errors.
  try contains risky code, catch handles errors,
  finally runs whether error occurs or not.

• **Question 2:** Why is error handling important in JavaScript applications?

- It prevents the program from crashing.
  It helps show user-friendly messages.
  Makes applications more stable and reliable.
  ```